

- b) An axisymmetric body with a linearly distributed load on the conical surfaces is shown in Fig. Q6 (b). Determine the equivalent point loads at nodes 6 and 4, the coordinate of the points 6 are (60, 40) mm point 4 are (40, 55) mm and point 2 are (20, 70) mm.

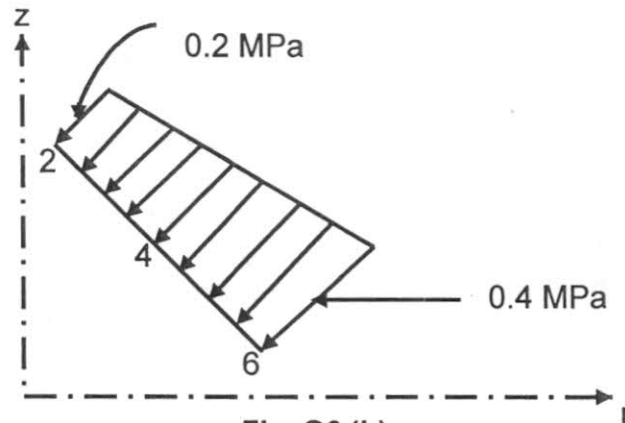


Fig. Q6 (b).

14 L3 3 1

Unit – IV

7. a) Write the shape function for four noded tetrahedral elements.
b) Write the shape function for Eight noded hexahedral elements.
c) Write the shape function for tetrahedral elements.
8. a) Derive the elemental stiffness matrix for the beam element.
b) For the beam shown in Fig. Q8 (b). determine the end reaction and deflection at mid-span take $E=200\text{GPa}$, $I=4\times 10^6\text{ mm}^4$

2 L3 4 1

8 L3 4 1

10 L3 4 1

10 L3 4 1

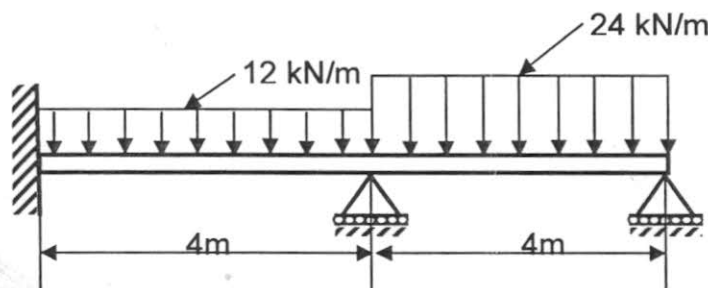


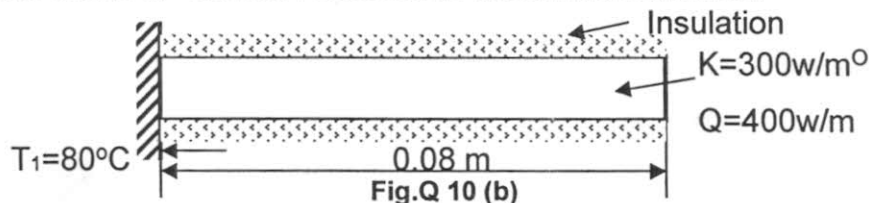
Fig. Q8 (b).

10 L3 4 1

Unit – V

9. a) Discuss the various steps involved in the finite element analysis of a one-dimensional heat transfer problem with reference to a straight uniform fin.
b) Determine the temperature distribution in a 1- Dimensional fin as shown in Fig Q 10 (b). There is a uniform generation of heat inside the wall of $Q=400\text{w/m}^3$. Diameter of the circular rod 0.02m.

10 L3 5 1



10 L3 5 1

10. a) Explain the characteristics of an algorithm
b) Explain the steps involved in designing an Algorithm.

10 L3 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
